SIMEON NORTH, JOHN HALL, AND MECHANIZED MANUFACTURING

To the Editor:

Historians generally agree that Simeon North and John Hall were pivotal figures in the introduction of machine tools and the attainment of interchangeability in American manufacturing technology; many of them accept the corollary that interchangeability of parts in manufactured mechanisms was achieved through the use of machine tools.[1] However, in two recent articles—one in this journal and one in *IA*—I have shown that interchangeability in small arms was realized at the national armories through improved skills in the handwork of artificers rather than with the precision of machinery.[2] I will show here that this also was true at the armories operated by Simeon North and John Hall.

Historical Background

Enthusiasm for the use of interchangeable parts in small arms originated among French engineers, scientists, and army officers in the late 18th century and was communicated to the U.S. Army by French officers serving in America.[3] Their influence, along with the obvious need to establish some measure of standardization in the disparate collection of artillery held by the army, led Secretary of War James McHenry in 1798 to establish uniformity as a goal for new American cannon.[4] At this time, "uniformity" meant dimensional uniformity and, for small arms, was expressed as a requirement for interchangeability of component parts. The ideal desired was that parts taken from any weapon would fit into, and function properly in, any other weapon of the same model. (Later, the definition of uniformity would

[1]Merritt Roe Smith, "John H. Hall, Simeon North, and the Milling Machine," *Technology and Culture* 14 (October 1973): 573–91, and *Harpers Ferry Armory and the New Technology* (Ithaca, N.Y., 1977), particularly chaps. 7 and 8; David A. Hounshell, *From the American System to Mass Production* (Baltimore, 1984), pp. 32–43.

[2]Robert B. Gordon, "Who Turned the Mechanical Ideal into Mechanical Reality?" *Technology and Culture* 29 (October 1988): 744–78, and "Material Evidence of the Manufacturing Methods Used in Armory Practice," *IA: Journal of the Society for Industrial Archeology* 14, no. 1 (1988): 22–35.

[3]Hounshell, pp. 25–28; Merritt Roe Smith, ed., *Military Enterprise and Technological Change* (Cambridge, Mass., 1985), p. 46.

[4]E. C. Ezell, "The Development of Artillery for the United States Land Service before 1861" (M.A. thesis, University of Delaware, 1963).

be broadened so as to include equivalence of physical properties, such as strength and hardness, in addition to uniformity of dimensions.)

An early attempt to introduce standardized small arms and interchangeable parts was made by Commissary-General Callender Irvine and in 1813 a contract with Simeon North for army pistols stipulated that the parts of the lock mechanisms be interchangeable. North had entered a career in manufacturing with the purchase of a sawmill on Spruce Brook in Berlin, Connecticut, which he converted to the manufacture of scythes in 1795. He obtained a contract for pistols from the U.S. government in 1799 and, with a succession of contracts, specialized in small-arms manufacture with a fine division of labor. To accommodate his growing arms business, he acquired a water privilege on the Coginchaug River in Middletown, Connecticut, on which he erected a three-story factory, 86 by 36 feet. The factories of two of the other three contractors, Robert Johnson and Nathan Starr, who figure in the subsequent introduction of milling technology, were located within a few miles of North's.

By 1816 North's factory, with its machinery and tools, was said to represent an investment of $100,000.[5] North enlarged the scope of his business in 1823 by becoming one of the contractors for the M1817 flintlock rifle.[6] In 1828 North undertook to make the M1819 breechloading (Hall) rifle under a contract that explicitly required interchangeability with the rifles of the same pattern being made at Harpers Ferry.[7]

Important developments in the mechanization of manufacturing took place at North's Middletown factory, as at other private armories and at John Hall's Harpers Ferry rifle works. M. R. Smith and Edwin Battison have brought forward documentary evidence showing that the first milling machine for cutting flat surfaces known to have been used in the United States, a machine described by Edward Parkhurst in 1900, was probably made at North's Middletown factory.[8] This machine was seen in 1851, still in use, but located in Robert Johnson's arms works. Smith has also found evidence that in 1829 Nathan Starr had a miller with a horizontal spindle supported on three bearings and fitted with cutters having the shape of the desired surface, and

[5]S. N. D. North and R. H. North, *Simeon North, First Official Pistol Maker of the United States* (Concord, N.H., 1913), chaps. 2–5.

[6]E. Norman Flayderman, *Flayderman's Guide to Antique American Firearms* (Northfield, Ill., 1977), p. 411.

[7]North and North, p. 184.

[8]Smith, "John H. Hall"; Edwin A. Battison, "A New Look at the 'Whitney' Milling Machine," *Technology and Culture* 14 (October 1973): 592–98; E. G. Parkhurst, "One of the Earliest Milling Machines," *American Machinist* 23 (1900): 217.

that in the same year William Smith and William Ferre, master machinists at North's works, built a miller in which the spindle could be adjusted vertically to regulate the depth of the cut.[9] Thus, the main elements of the milling machine that would later be made commercially under the name "Lincoln" and widely used by armories and other industries had been developed at three arms factories within a few miles of each other between 1816 and 1830. These developments in milling technology would have been known to John H. Hall as he built new metal-cutting machines for the manufacture of his breech-loading rifle at the Harpers Ferry Armory.[10] Because the Patent Office fire of 1836 destroyed the drawings of Hall's machines and the demolition of the rifle works in 1861 destroyed the machines themselves, only a limited description of them can be pieced together from surviving documentary evidence.

The best (but not very explicit) account of Hall's machinery is the report on the rifle works prepared by James Carrington, Luther Sage, and James Bell in 1827.[11] They stress the unusually heavy and accurate construction of the machines they inspected, an account echoed by Fitch, who describes them as "excessively solid and heavy."[12] Smith has shown from a study of Harpers Ferry purchase records that many of Hall's machines were built on cast-iron frames.[13] Although this was not new technology—the Portsmouth pulley-block machinery and the milling machines used in the Connecticut armories were built on cast-iron frames[14]—the use of such frames undoubtedly helped Hall counter the lack of structural rigidity that had been a serious limitation of earlier machine tools. These machines, together with those made at Middletown, appear to be the first milling machines of substantial construction made for production work on metal parts. To find how they were used we need to turn to industrial archaeology.

Evidence from Industrial Archaeology

In 1987, I examined all the Hall-type breech-loading arms in the collection of the Springfield Armory Museum for evidence of man-

[9]Smith, "John H. Hall."

[10]Ibid.

[11]Stephen V. Benet, ed., *A Collection of Annual Reports and Other Important Papers* (Washington, D.C., 1878), pp. 153–67; Smith, *Harpers Ferry* (n. 1 above), pp. 200–208.

[12]Charles H. Fitch, "Report on the Manufactures of Interchangeable Mechanism," in *Tenth Census of the United States (1880)*, vol. 2, *Report on the Manufactures of the United States* (Washington, D.C., 1883), p. 25.

[13]Smith, *Harpers Ferry*, p. 228.

[14]Carolyn C. Cooper, "The Portsmouth System of Manufacture," *Technology and Culture* 25 (April 1984): 182–225; Robert S. Woodbury, *Studies in the History of Machine Tools* (Cambridge, Mass., 1972), p. 20.

ufacturing methods.[15] The two M1819 rifles in the best condition, one made at Harpers Ferry in 1838 and one at Middletown in 1832, were disassembled, cleaned, and scrutinized. These arms require a close fit between the barrel and the breech block because Hall's design includes no positive gas seal; rather, Hall minimized gas leakage due to elastic deflection of the breech mechanism under the force of the discharge by placing the locking lugs at the front of the breech block.[16] The bearing surfaces of the breech, breech-block face, locking lugs, locking keys, and the slots in the receiver into which the keys fit all have a cylindrical shape; the fits between these surfaces, which must be free-running, determine the clearance between the barrel and breech block. Making this breech mechanism was probably the most difficult precision-manufacturing task attempted in the United States before the mid-19th century.

Interchangeability

Nineteenth-century writers sometimes made claims for the attainment of interchangeability in small arms at dates before 1848, claims since found, as in the case of the Whitney Armory, to be inaccurate.[17] Contemporary documents are not usually explicit about how interchangeability was demonstrated. North's 1813 pistol contract stipulates that "the component parts of pistols, are to correspond so exactly that any limb or part of one Pistol, may be fitted to any other Pistol."[18] The report made on North's armory by James Stubblefield and Roswell Lee in 1816 implies that this was attained: "He [North] has made an improvement in the lock by fitting every part to the same lock."[19] To confirm this, in 1987 James B. Smith tested the interchangeability of the locks in the stocks of the two M1811, two M1813, and three M1816 North pistols in his collection; none of the locks would interchange with their mates of the same model. Thus, it appears that the 1813 contract was expressing an ideal that was not attained in practice.

The documentary record of progress made in the next decade by Simeon North and John Hall toward realizing practical interchangeability also lacks specificity. The report of the Carrington committee states that tests made for the committee demonstrated that the stocks

[15]The techniques used are described in Gordon, "Material Evidence" (n. 2 above).

[16]A technical description is given by David F. Butler, *United States Firearms, the First Century, 1776–1875* (New York, 1971), pp. 132–36; I use his terminology for the parts of the rifle.

[17]Robert S. Woodbury, "The Legend of Eli Whitney and Interchangeable Parts," *Technology and Culture* 1 (Summer 1960): 235–53.

[18]North and North (n. 5 above), p. 81.

[19]Ibid., p. 105.

of M1819 rifles were interchangeable.[20] Yet, while indicating that the receivers were disassembled, it does not say that they were successfully reassembled from the mixed parts. A more explicit statement was made by the Chief of Ordnance in 1827 when he declared the M1819 rifles made at Harpers Ferry to be interchangeable.[21] North's 1828 contract for M1819 rifles stipulated that they be interchangeable with those made at Harpers Ferry, "said exchanges [of parts] to be made without impairing in the least the efficiency or perfection of the Arms which are thus composed of exchanging parts."[22] This contract, unlike the 1813 one, was enforced; with the aid of gages of his own making, North satisfied Hall that his rifles were interchangeable with those made at the Harpers Ferry rifle works.[23]

To test how good the interchangeability achieved by Hall and North was, Stuart Vogt, curator at the Springfield Armory Museum, tested the two M1819 rifles described above. He found that both will function with their breech blocks interchanged, but with some loss in the ease of operating the mechanism. In order to get a quantitative indication of the uniformity achieved by Hall and North, I measured eleven critical dimensions of the breech mechanisms of these two rifles. The clearance between the breech block and barrel (the head space) measured with feeler gages was found to be 0.009 inches on the Harpers Ferry rifle and 0.006 inches on the North rifle. (Butler states that the Hall rifles in "new" condition in the Winchester Gun Museum had head spaces of 0.002 to 0.004 inches.)[24] The average deviation of the other ten dimensions between the two rifles was 0.0027 inches. This may be compared with an average deviation of 0.0042 inches in lock parts made at the Springfield Armory at this time.[25] These results support the documentary evidence of interchangeability and show that both Hall and North had achieved a higher standard of precision in manufacturing than had the national armories, at least through the 1830s.

Use of Machine Tools

The tool marks found on the two rifles studied reveal the manufacturing technology used to achieve gage dimensions. Clear evidence of

[20]Benet, ed. (n. 11 above).

[21]Chief of Ordnance to the Secretary of War, 1827, quoted in Thales L. Ames, "Captain John H. Hall: His Contribution to the Art of Arms," *Army Ordnance* 3 (1923): 346–49.

[22]North and North, p. 160.

[23]Smith, *Harpers Ferry* (n. 1 above), p. 211.

[24]Butler, p. 139.

[25]Gordon, "Who Turned the Mechanical Ideal" (n. 2 above).

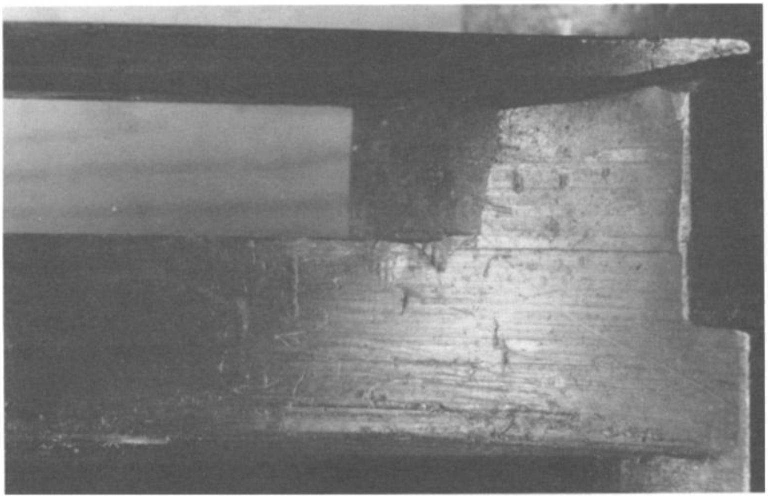

FIG. 1.—View of the bottom of the breech block of a U.S. Model 1819 rifle made at Harpers Ferry in 1838, showing the cavity into which the lock parts fit. Striations accurately parallel to each other from two milling cuts are visible. "Carrot marks" on these striations show that the cutter was rotating toward the edge visible in the picture. The short, transverse marks on the milled surfaces are tears, which form easily at slag inclusions in wrought iron. Random, diagonal marks appear to be gouges left by a file or scraper used to remove burrs after milling. The absence of eccentricity marks suggests that a plunge cut was used; the absence of chatter marks, that the milling machine was of very rigid construction. The surface shown does not mate with any other surface in the mechanism. (The scale is shown by the width of the breech block, 1 inch. For methods of interpretation, see n. 2 above.)

milling was found on the inside of the breech block of the Harpers Ferry rifle (fig. 1), showing that a milling machine was used to cut out the cavity that accommodates the lock parts. The only other surface carrying milling marks was the inside of the pan; all other surfaces in the rifle mechanism were filed to their final dimensions. The same pattern of tool marks was found on the Harpers Ferry and North rifles. Figure 2 shows the intersecting striations characteristic of filing found on the side of the breech block. In addition to the flat surfaces, the cylindrical surfaces of the locking lugs, keys, and key slots were all filed. Since the bearing surfaces of the locking lugs are concave, a specially curved file would have been required for this work. Flat filed surfaces were found to be accurately flat when tested against a standard straightedge, and the cylindrical surfaces were found to mate well with each other. The pattern of filing is identical except for small details on the Harpers Ferry and North rifles; the standard of workmanship on both is very high. The same evidence of filing was found on all the rifles and carbines with the Hall breech mechanism in the Springfield Armory

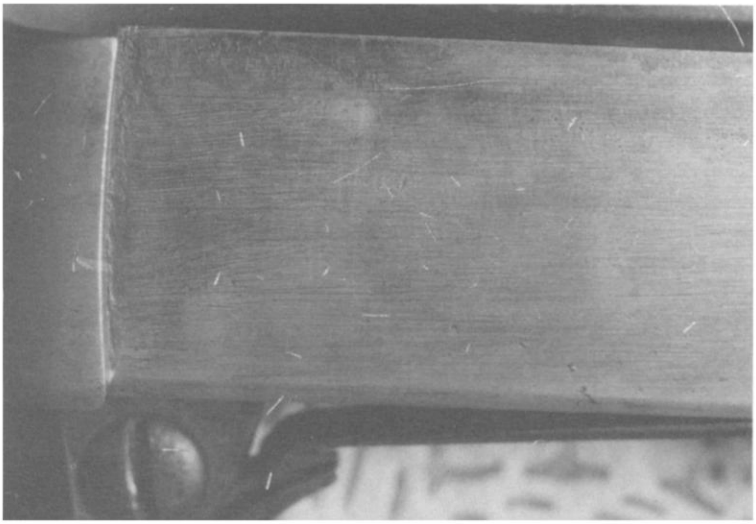

FIG. 2.—Side view of the breech block of a U.S. Model 1819 rifle made at Simeon North's Middletown, Connecticut, armory in 1832. The locking lug, which is on the forward end of the breech block, is visible. The flat surface of the breech block is covered with the intersecting striations characteristic of hand filing. The file strokes were taken along the length of the block and, since they come right up to the lug, were made by draw filing. Striations formed by hand filing are present on the curved surfaces of the locking lug. (The scale is shown at the height of the breech block, 1 inch.)

collection. Thus, the evidence shows that the manufacturing technology did not change significantly through the end of production of these arms in 1853, that a very high standard of filing was attained by the artificers at both the Harpers Ferry rifle works and North's armory, and that Hall and North had designed effective gaging systems before 1832 for the assurance of interchangeability.

Since all the surfaces that had to be brought to gage dimensions in these rifles were hand filed, it follows that the machine tools used in making these parts were not capable of the accuracy needed to attain a close fit of the breech mechanism and interchangeability of parts. It might be argued that these parts were machined to size and then simply cleaned up with a file, leaving the marks observed. As anyone who has tried it will testify, however, it is exceedingly difficult to file either a flat or a cylindrical surface; if such surfaces were already made to correct dimension by machine, no mechanic would touch them with a file.

We may conclude from these observations that machine tools were used by both Hall and North to rough out forged parts which were then filed to gage by hand with the aid of filing jigs.[26] To attain the dimensional accuracy shown by our measurements, a very high level of skill in handling both files and gages was required of the artificers. In my recent *T&C* article I have argued that the introduction of milling technology greatly reduced the physical labor of removing metal from forged gun parts, thereby allowing artificers to concentrate on the finer work of bringing parts to final dimensions.[27] New archaeological evidence from the sites of North's two factories supports this interpretation. In an exploration of North's Berlin factory site by James B. Smith and others, an uncut lock plate forging for an M1811 pistol was recovered. Measurement shows this to be 3/16-inch oversize on its periphery and up to 9/16-inch thicker than the finished plate. The excess metal would need to have been cut off the forging by heavy hand labor or by machine tools. The amount of waterpower potential available at the Berlin factory site can be calculated from the drainage basin characteristics and the height of the dam.[28] The factory could accommodate about forty artificers in 1809,[29] and the power potential per worker would then have been only about 0.04 horsepower. (For comparison, a man winding a winch for an eight-hour day can generate about 0.08 horsepower.) With such a small amount of power available at the Berlin shop, heavy handwork, chipping and filing the forged gun parts, would have taken up much of the working time of North's artificers. This was undoubtedly the reason North acquired the new factory site on the Coginchaug River. Since most of the waterpower system at this site is still in place, the power available there can be calculated. About fifty artificers were at work at the Middletown factory in 1816,[30] and the power potential was 1.9 horsepower per man, sufficient to provide for a substantial reduction of the physical labor required of the artificers and to undertake the heavy machining needed to make the lock parts for the M1819 rifle.

These conclusions about the use of machine tools in making the M1819 rifle are at variance with the interpretation of the report of the Carrington committee favored by many historians. The report describes demonstrations showing that metal parts for the M1819 rifle

[26]This was also the practice at other armories throughout the 19th century, see ibid.

[27]Ibid.

[28]Potential power is the power of the fall and mean flow of water at the site. The actual power available will be less, depending on the efficiency of the power system used, and will vary as the flow of water varies. See Robert B. Gordon, "Hydrological Science and the Development of Waterpower for Manufacturing," *Technology and Culture* 26 (April 1985): 204–35. I estimated the height of the dam at North's Berlin shop during an inspection of the site.

[29]North and North (n. 5 above), p. 68.

[30]Roswell Lee and James Stubblefield, quoted in ibid., p. 105.

could be made at lower cost and more accurately by Hall's machinery than by experienced artificers working with files. It states: "Arms have never been made . . . so that those parts, on being changed, would equally well suit when applied to every other arm. But the machines we have examined effect this with a certainty."[31] But elsewhere the report always speaks of interchangeability being attained with the aid of the machines or by Hall's system of manufacture. It never says that filing to gage was not required. When some allowance is made for the committee members' obvious enthusiasm for the machinery they saw, it seems to me that their report is not at variance with the archaeological evidence.

More than two decades after North and Hall were using milling machines in the production of the M1819 rifle, the Springfield Armory had not succeeded to a comparable degree in using machinery for the production of small-arms components. Since information about developments in manufacturing technology was exchanged rapidly among armories at this time, technical or economic factors must have been responsible for the delay, and several of these can be identified. Some parts in the Hall breech mechanism required much more rough cutting of the forgings than was needed on other arms; the cavity for the lock mechanism in the breech block would have been particularly difficult to make with hand tools. The performance of Hall's machinery would have been limited by the carbon-steel milling cutters then available. These cutters had small, file-like teeth,[32] and heavy cuts would have been impossible because of inadequate clearance for the chips and rapid dulling of the cutting edges. Either resharpening or making new cutters would have been costly, particularly in the absence of machines for cutter grinding. A test done for the Carrington committee showed that the work performed by the machines could be done by hand in only one-third more time[33] (the difference would have been less on arms simpler than the M1819 rifle). Filing to gage was still required after machining. It would not have been economic for other armories, where less rough cutting of forgings was needed, to invest in specialized machine tools that probably could not achieve a sufficient reduction of labor to justify their initial cost. A reason that applies specifically to the Springfield Armory is the relatively small power potential available there, less than one horsepower per production worker in 1816, about half that available at

[31]Benet, ed. (n. 11 above).

[32]A cutter of this type is illustrated in C. Fremont, *Files and Filing* (London, 1920), p. 22.

[33]Benet, ed.

North's armory at the time milling technology was being developed there.[34]

The archaeological evidence presented here confirms the development of milling technology at the armories of Simeon North and John Hall in the years before 1832 and shows the importance of these machine tools in the manufacture of the Hall rifle. It also shows that the high level of precision achieved at these armories cannot be credited directly to the use of machine tools but must be ascribed to the development of efficient gaging systems and the very high standard of hand filing attained by the artificers at both armories.

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[34]This calculation is based on the drainage basin characteristics, the description of the Water Shops in Derwent S. Whittlesey, "A History of the Springfield Armory" (MS, Springfield Armory National Historic Site, Springfield, Mass., 1920), and employment data in Felicia J. Deyrup, *Arms Makers of the Connecticut Valley* (Northampton, Mass., 1948), p. 245. The archaeological work necessary for making similar power estimates at the Harpers Ferry Rifle Works has not yet been done but, in view of the flow available there, it was certainly much greater.